AWS Storage Services: Buckets, Tables, and Vector Storage

1. AWS Storage Buckets (Amazon S3)

Overview

Amazon Simple Storage Service (Amazon S3) is AWS's object storage service designed for scalability, durability, and high availability. Data is stored as **objects** within **buckets**, making S3 suitable for unstructured and semi-structured data.

Key Characteristics

- **Storage type:** Object storage
- **Scalability:** Virtually unlimited
- **Durability:** 99.999999999% (11 nines)
- **Availability:** High availability across AWS Regions
- **Consistency:** Strong read-after-write consistency

Common Use Cases

- Data lakes and analytics storage
- Backup and archival storage
- Static website hosting
- Media storage (images, videos, documents)
- ML dataset storage

Storage Classes

- S3 Standard
- S3 Intelligent-Tiering
- S3 Standard-IA / One Zone-IA
- S3 Glacier Instant Retrieval
- S3 Glacier Flexible Retrieval
- S3 Glacier Deep Archive

Security & Management

- IAM-based access control
- Bucket policies and ACLs
- Server-side and client-side encryption
- Versioning and lifecycle policies

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2. AWS Storage Tables (Amazon DynamoDB)

Overview

Amazon DynamoDB is a **fully managed NoSQL key-value and document database** service. Data is stored in **tables**, each containing items identified by primary keys.

Key Characteristics

- **Storage type:** NoSQL (Key-Value / Document)
- **Latency:** Single-digit millisecond
- **Scalability:** Automatic, serverless scaling
- **Availability:** Multi-AZ replication by default

Data Model

- **Tables:** Collection of items
- **Items:** Individual records
- **Attributes:** Key-value pairs
- **Primary Keys:** Partition key or Partition + Sort key

Common Use Cases

- High-throughput transactional systems
- User profiles and session management
- IoT telemetry storage
- Gaming leaderboards
- Real-time applications

Performance & Capacity

- On-demand or provisioned capacity modes
- Global Secondary Indexes (GSI)
- Local Secondary Indexes (LSI)
- DynamoDB Accelerator (DAX) for caching

Security & Reliability

- Fine-grained IAM permissions
- Encryption at rest and in transit
- Point-in-time recovery
- Global Tables for multi-region replication

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3. AWS Vector Storage

Overview

AWS does not offer a single standalone "vector database," but provides **multiple services that support vector storage and similarity search**, commonly used in AI, ML, and GenAI workloads.

Vector storage is used to store **embeddings** (high-dimensional numerical vectors) and perform **similarity search** using distance metrics such as cosine similarity or Euclidean distance.

Primary AWS Options for Vector Storage

3.1 Amazon OpenSearch Service (Vector Engine)

- Native support for vector fields
- Approximate Nearest Neighbor (k-NN) search
- Scales horizontally
- Commonly used for semantic search and RAG pipelines

Use cases:

- Semantic search
- Recommendation systems
- Retrieval-Augmented Generation (RAG)

3.2 Amazon Aurora / RDS with pgvector

- PostgreSQL extension (pgvector) support
- Stores vectors as columns
- Suitable for structured + vector hybrid queries

Use cases:

- Applications combining relational data with embeddings
- Moderate-scale similarity search

3.3 Amazon S3 + Custom Vector Index

- Vectors stored in S3
- Index built using FAISS or similar libraries
- Typically used with SageMaker or EC2

Use cases:

- Large-scale offline embedding storage
- Batch similarity search
- Research and experimentation

Integration with AWS AI Services

- Amazon SageMaker for embedding generation
- Amazon Bedrock for foundation model embeddings
- Lambda and ECS/EKS for inference pipelines

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4. Comparative Summary

Feature	**S3 Buckets**	**DynamoDB Tables**	**AWS Vector Storage**
Data Type	Objects (files)	Key-value / JSON	Numerical vectors
Query Style	Object retrieval	Key-based queries	Similarity search
Scalability	Virtually unlimited	Automatic	Service-dependent
Typical Workloads	Storage, backup, analytics	Transactions, real-time apps	AI/ML, GenAI, search

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5. Conclusion

AWS provides diverse storage solutions tailored to different data and workload requirements. **S3 buckets** are ideal for durable and scalable object storage, **DynamoDB tables** support high-performance NoSQL workloads, and **AWS vector storage solutions** enable modern AI-driven applications such as semantic search and generative AI systems. Selecting the right service depends on data structure, access patterns, latency requirements, and scalability needs.